

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
Prof. Dr. Sofia C. Olhede

Practice Exam — Week 2: ARMA Models and the PACF

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

(a) State the definition of white noise and write down its ACVS γ_τ and ACF ρ_τ . Explain in one sentence why “uncorrelated” is weaker than “independent”. (1.5 pts)

(b) Consider the MA(2) process

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}.$$

Compute the mean and the autocovariance γ_τ for *all* $\tau \in \mathbb{Z}$, and write down ρ_1 and ρ_2 . (2.5 pts)

(c) For which lags τ does the ACVS of an MA(q) vanish? State the general rule and explain how it is used to read off the order q from an estimated ACF. (1 pt)

Problem 2 (5 points)

Consider the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$.

(a) State the condition on ϕ for the process to be causal, and explain it in terms of the root of the AR polynomial $\Phi(z) = 1 - \phi z$. (1 pt)

(b) Assuming $|\phi| < 1$, derive the causal MA(∞) representation $X_t = \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$ by iterating the recursion. Give the ψ -weights and verify $\sum_k |\psi_k| < \infty$. (2 pts)

(c) Using the representation in (b), show that

$$\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

Why does this ACF *tail off* rather than cut off, and what does that say about identifying p from the ACF? (2 pts)

Problem 3 (5 points)

(a) Consider the AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ with $\phi_1 = \frac{1}{2}$, $\phi_2 = -\frac{1}{4}$. Using the causality conditions $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$, verify that the process is causal. (1 pt)

(b) From the Yule–Walker equations $\rho_\tau = \phi_1 \rho_{\tau-1} + \phi_2 \rho_{\tau-2}$ (with $\rho_0 = 1$, $\rho_{-1} = \rho_1$), compute ρ_1, ρ_2, ρ_3 . (2 pts)

(c) Using the Durbin–Levinson recursion

$$\alpha_{1,1} = \rho_1, \quad \alpha_{\tau,\tau} = \frac{\rho_\tau - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_j}, \quad \alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau} \alpha_{\tau-1,\tau-j},$$

compute $\alpha_1, \alpha_2, \alpha_3$ from the values in (b). Read off the implied model order and state the ACF/PACF identification rule it illustrates. (2 pts)

Problem 4 (5 points)

(Revision of Week 1: stationarity, the ACVS & ergodicity.) Let $\{Y_t\}$ be i.i.d. with mean 0 and variance σ_Y^2 , and define $X_t = Y_t + \frac{1}{2}Y_{t-1}$.

(a) Compute $\mathbb{E}[X_t]$ and the autocovariance γ_τ^X for all $\tau \in \mathbb{Z}$, give the lag-1 autocorrelation ρ_1 , and explain in one sentence why $\{X_t\}$ is weakly stationary. (2 pts)

(b) (*Validity via positive semi-definiteness.*) A candidate sequence is a valid ACVS of some stationary process only if it is positive semi-definite. Decide whether

$$\gamma_0 = 1, \quad \gamma_{\pm 1} = 0.8, \quad \gamma_\tau = 0 \quad (|\tau| \geq 2)$$

is a valid ACVS, by forming the 3×3 covariance matrix $\Gamma_3 = [\gamma_{j-k}]_{1 \leq j, k \leq 3}$ and testing its definiteness (e.g. via its determinant). (2 pts)

(c) (*Mean ergodicity.*) Show that if $\sum_{\tau \in \mathbb{Z}} |\gamma_\tau| < \infty$ then $\text{Var}(\bar{X}_n) \rightarrow 0$ as $n \rightarrow \infty$, where $\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t$, so that $\bar{X}_n$ is a consistent (mean-ergodic) estimator of $\mu = \mathbb{E}[X_t]$. (1 pt)

